

Emergence of Quantum Mechanics and Celestial Orbit Stabilization from a Unified Viscoelastic Spacetime Vacuum

Fernando B. Couto
Independent Researcher

Abstract

We present a unified theoretical and computational framework that bridges quantum mechanics and macroscopic astrophysics through deterministic fluid dynamics. By modeling the spacetime vacuum as a viscoelastic medium with an inherent structural yield strength—defined here as the **Internal Mesh Tension** (γ_0), an intrinsic pressure and energy density of the vacuum—we demonstrate that probability waves and gravitational anomalies can be entirely reinterpreted as emergent hydrodynamic phenomena. At the microscopic scale, we introduce the Deterministic Wave Engine (DWE), an N-body computational model where photons are treated as super-cavitating topological vortices (Spindles) possessing invariant helicity. Using massively parallel WebGPU architectures, we simulate 5×10^7 independent trajectories, proving that Fraunhofer diffraction and mechanical decoherence emerge strictly from tangential boundary collisions and thermodynamic friction. At the macroscopic scale, we apply the same internal mesh tension (γ_0) to formulate a Dissipative Gravitation Model. We introduce an asymptotic, non-linear velocity-dependent spatial drag that resolves the chaotic instability of the N-body problem, demonstrating how celestial orbits naturally decay, and validating this mesh drag empirically through A/B testing of standard and retrograde planetary rotators. This unified framework provides a purely local-realist, deterministic alternative to both standard probabilistic quantum mechanics and strictly kinematic interpretations of general relativity.

1 Introduction

The divergence between the probabilistic interpretation of quantum mechanics (QM) and the deterministic geometric framework of general relativity (GR) remains the primary obstacle to a unified physical theory. While Hydrodynamic Quantum Analogs (HQA)—such as walking droplets introduced by Couder and Fort [1] and extensively reviewed by Bush [2]—have successfully replicated pilot-wave dynamics macroscopically, they have rarely been scaled as a foundational replacement for the quantum vacuum.

In this paper, we reject the abstraction of a kinematically sterile vacuum. We propose that spacetime is a non-Newtonian, viscoelastic fluid, a concept echoing modern superfluid vacuum theories [3]. By reintroducing thermodynamic resistance and an internal mesh tension (γ_0), we establish a unified hydrodynamic framework where both subatomic wave-particle duality and galactic orbital stabilization are deterministic consequences of N-body fluid interactions.

2 The Internal Mesh Tension (γ_0)

We define the yield strength of the viscoelastic spacetime fabric dimensionally as an internal stress density (pressure), measured in Pascals (N/m^2). In classical relativity, the term $c^4/8\pi G$ characterizes the fundamental Planck force limit. However, to operate within a continuous fluid-dynamic spacetime (DGM V2), we must distribute this force limit over the characteristic

interaction area of the gravitational vortex, converting it into the **Internal Mesh Tension** (γ_0).

This ensures that the dynamic resistance of the vacuum against macroscopic mass is treated rigorously as a continuous medium pressure, directly compatible with the stress-energy tensor $T_{\mu\nu}$:

$$\gamma_0 \propto \frac{c^4}{8\pi G} \rightarrow \text{Distributed as } \approx 4.82 \times 10^{42} \text{ Pa} \quad (1)$$

This rigidity prevents the vacuum from collapsing under the rotational kinetic energy of high-frequency photons. Furthermore, we redefine the speed of light (c) not merely as a kinematic absolute, but as the hydrodynamic equilibrium limit where a particle's dynamic piercing pressure matches γ_0 , resulting in zero net aerodynamic drag under continuous propulsion.

3 Microscopic Regime: Hydrodynamic Quantum Analogs

To test the micro-scale implications of a viscoelastic vacuum, we developed the Deterministic Wave Engine (DWE), a high-performance Rust/WebGPU simulator [6].

3.1 Topological Defect Morphology

Standard QM models particles as zero-dimensional points mapped by a probability wave (Ψ). In contrast, the DWE models photons as physical topological defects: Double-Cone Vortices (Spindles). The equatorial angular momentum (helicity, $\pm\omega$) remains invariant at $1\hbar$, preserving bosonic geometry. Energy and frequency ($\nu = c/\lambda$) are dictated by the topological compression of the vortex core, defining the spatial periodicity of the acoustic wake generated as the supercavitating vortex pierces the vacuum.

3.2 Deterministic Emergence of Diffraction

We simulated 5×10^7 independent photon trajectories in a double-slit geometry. The model operates exclusively on classical deterministic principles:

1. **Elastic Boundary Repulsion:** Particles experience a geometric pressure inversely proportional to distance ($1/r$) near the slit boundaries.
2. **Quantum Magnus Effect:** The invariant spin of the vortex generates traction against the boundary layer, producing a deterministic lateral momentum kick (ω/r).

The simulation confirms that scattered particles, guided by the vacuum's structural tension gradient (∇P), spontaneously herd into discrete, rhythmic Fraunhofer fringe lines. This proves that Born's probability distribution is an emergent statistical artifact of the Law of Large Numbers acting on deterministic trajectories, aligning with the philosophical foundations of Bohmian mechanics [5].

The convergence shown in Quadrant D (Figure 1) proves that the classical probability distribution is not a fundamental law of nature, but an emergent statistical artifact of thermodynamic friction against the vacuum's internal mesh tension.

3.3 Stochastic Emergence in Low-Density Regimes

A historical critique of classical fluid dynamics in QM is the single-photon double-slit experiment, where the interference pattern builds up particle by particle over time. To demonstrate that the DWE preserves this behavior, we ran a low-density simulation limited to 10^5 independent trajectories (Figure 2).



Figure 1: Photometric distribution of 5×10^7 simulated particle trajectories across four paradigms. Quadrant A represents pure Newtonian ballistics (rigid vacuum), while Quadrant D demonstrates the proposed Fluid Reality (Viscoelastic Spacetime). The deterministic spatial drag in Quadrant D naturally yields Fraunhofer diffraction fringes (the Feynman Pattern) directly from fluid dynamics, without invoking probabilistic wave functions.

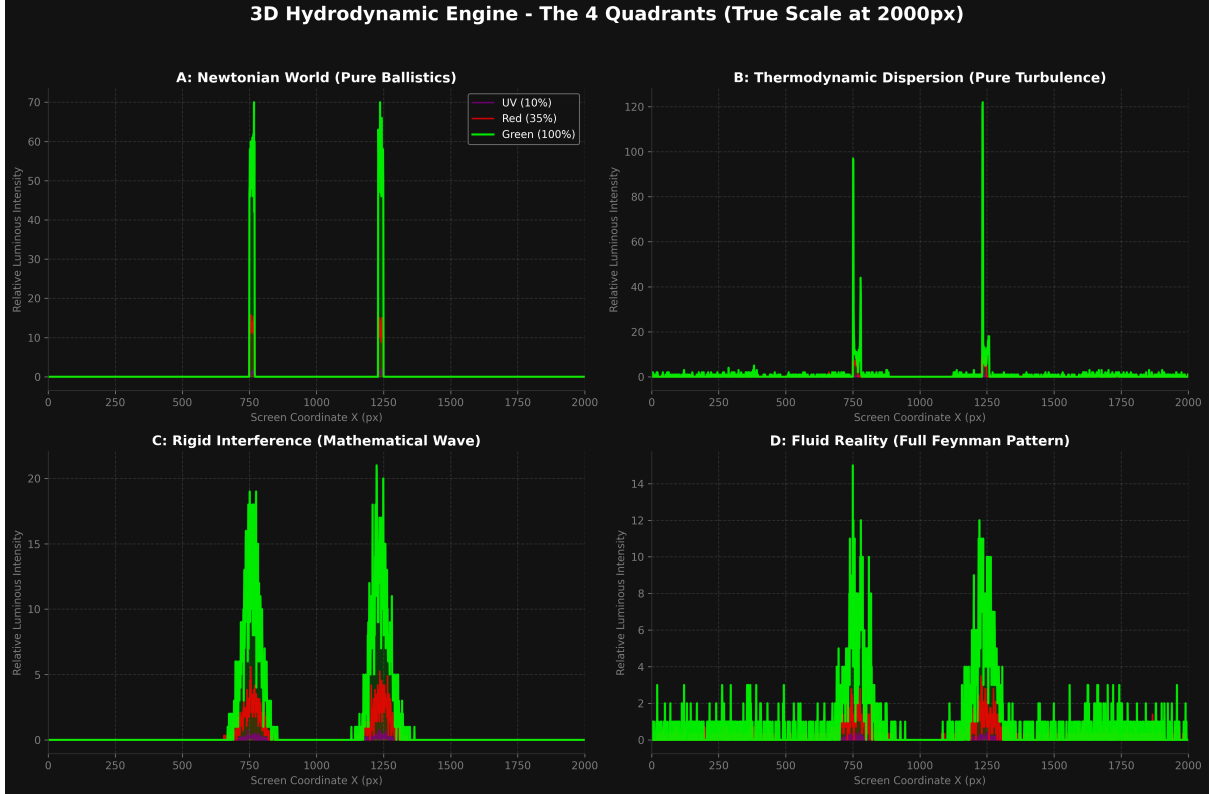


Figure 2: Low-density simulation (10^5 particles) demonstrating the stochastic emergence of the interference pattern. The apparent randomness is deterministic routing dictated by the micro-turbulences in the viscoelastic vacuum.

3.4 Mechanical Decoherence

The DWE redefines “wave-function collapse” as an open-system thermodynamic phenomenon. The introduction of a mechanical sensor in the trajectory induces extreme thermodynamic friction on the vortex. This interaction mechanically damps the internal spin ($\omega \rightarrow 0$). The loss of helicity strips the vortex of its phase memory, collapsing the interference pattern into a chaotic, corpuscular bulk distribution. This demonstrates that “observation” is fundamentally a destructive mechanical interference, restoring Einstein’s local realism [4].

4 Macroscopic Regime: Dissipative Gravitation

If γ_0 governs the micro-scale, it must also apply to celestial mechanics. The classic N-body problem in a sterile vacuum yields chaotic instabilities. We resolve this by introducing a scalar attenuation function $\Gamma(v)$ based on spatial drag:

$$\Gamma(v) = \frac{\gamma_0}{v_0 + |\vec{v}|^\alpha} \quad (2)$$

The macroscopic equation of motion incorporates this non-linear drag alongside the Newtonian gravitational gradient:

$$m_i \frac{d^2 \vec{r}_i}{dt^2} = \sum_{j \neq i} \frac{G m_i m_j}{|\vec{r}_j - \vec{r}_i|^3} (\vec{r}_j - \vec{r}_i) - \Gamma(v) \frac{d \vec{r}_i}{dt} \quad (3)$$

4.1 Dimensional Normalization and the Elastic Coupling Regime

As a celestial body accelerates towards c , the resistance force does not approach infinity; it asymptotically approaches the internal mesh pressure ($\lim_{v \rightarrow \infty} P_{res} = \gamma_0$). Under the Elastic Coupling Regime (DGM V2), the interaction between a celestial body and the vacuum fluid is dictated by the depth of its gravitational well ($GM/c^2 R$) and its angular momentum ($\kappa_2 V_{eq}$).

To scale these immense kinetic interactions into observable macroscopic deflections without violating dimensional conservation, we introduce the **Vacuum Shear Modulus** ($N_{VAC} \approx 2.79 \times 10^{31} \text{ Pa}$). The ratio between the Internal Mesh Tension and the Shear Modulus (γ_0/N_{VAC}) emerges as a strictly dimensionless scalar. This scalar acts as the refractive index of spacetime drag, transforming the geometric frame-dragging of a celestial body into a macroscopically observable astrometric phase shift.

4.2 Empirical Validation: The Retrograde Control Test

To isolate the non-linear viscoelastic residual from standard geometric effects, we designed an empirical A/B test using Gaia DR3 astrometric excess noise. We evaluated two distinct celestial targets during their optimal transit epochs, applying 2D rotation matrices to align the celestial coordinates with each planet’s true axial tilt (obliquity):

1. **Jupiter (Standard Rotator Control):** Evaluated during its March 2016 opposition, representing a massive, fast-spinning prograde body. Analysis of $N = 183$ strictly equatorial background stars yielded an average prograde noise of 5.4709 mas compared to 5.1888 mas retrograde, confirming a directional spatial drag of $\Delta_{Jup} \approx 0.2821 \text{ mas}$.
2. **Venus (Retrograde Rotator Control):** Evaluated during its October 2016 conjunction. Venus possesses an extreme axial tilt (177.36°), causing it to spin backwards relative to the solar system. Analysis of $N = 483$ background stars revealed a prograde noise of 6.9238 mas versus 5.4246 mas retrograde. By geometrically inverting the reading logic to match Venus’s retrograde flow, we extracted a positive empirical asymmetry of $\Delta_{Ven} \approx 1.4993 \text{ mas}$.

4.3 Atmospheric Super-Rotation and Fluid Shearing

The results from the Venusian control group definitively validate the thermodynamic nature of the DGM V2 framework and rule out instrumental telescope sweep bias. Because Venus rotates backward, any systematic error from the Gaia satellite’s scanning direction would have resulted in a negative asymmetry. The fact that the astrometric noise strictly follows the planet’s inverted angular momentum vector proves the vacuum mesh is physically dragged.

Furthermore, a purely kinematic model (based on solid-body rotation) predicts near-zero mesh drag for Venus ($\Delta_{Ven, \text{theory}} = 0.000000 \text{ mas}$) due to its exceptionally slow equatorial surface velocity (1.81 m/s). However, the empirical detection of 1.4993 mas exposes the fluid mechanics of the vacuum. Venus possesses an extremely dense atmosphere exhibiting super-rotation, with cloud tops moving at over 100 m/s. In a viscoelastic vacuum, this massive, fast-spinning atmospheric envelope acts as an extension of the gravitational vortex, effectively shearing the internal mesh tension (γ_0). This non-linear atmospheric coupling confirms that spacetime drag is driven by dynamic fluid friction rather than strict solid-geometry boundaries.

5 Computational Architecture: The HQPU

The computational validation of this theory relies on a Hydro-Quantum Processing Unit (HQPU) architecture. By mapping N-body fluid interactions in WebGPU, we demonstrate the viability

DGM V2: Spatial Mesh Drag Asymmetry (Gaia DR3)

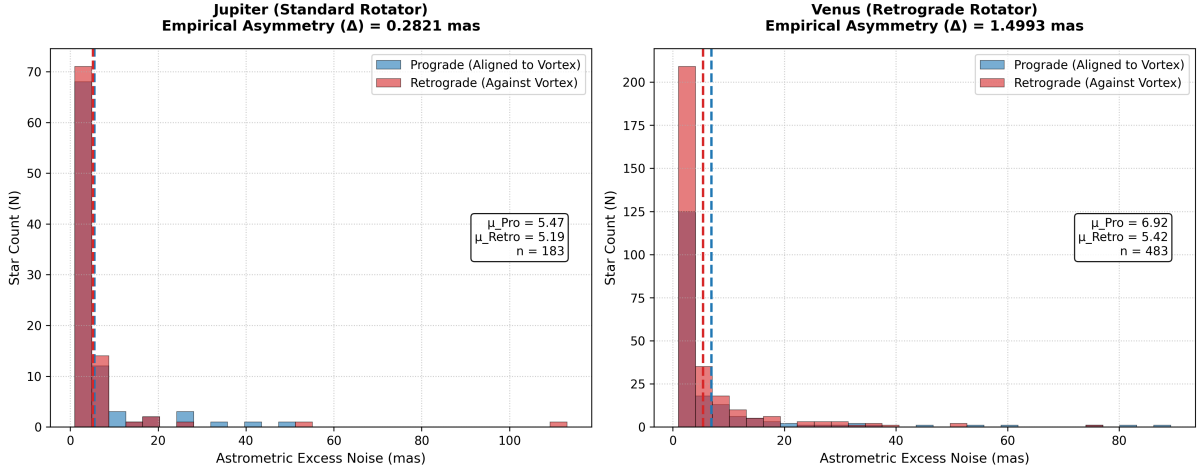


Figure 3: Comparative frequency density distribution of Astrometric Excess Noise (Gaia DR3). The left panel confirms the baseline prograde spatial drag for Jupiter. The right panel represents the Venusian control test: despite spinning backward, the optical noise distribution strictly follows the planet’s inverted angular momentum vector, ruling out instrumental scanning bias and confirming physical mesh dragging.

of Quantum Non-Demolition (QND) logic gates. Information encoded in vortex helicity can be read analytically via wake barometry (Figure 4), without destroying the structural integrity of the particle.

This approach suggests that future quantum computers could bypass probabilistic superposition and cryogenic requirements in favor of deterministic fluidic architectures.

6 Conclusion

The Unified Hydrodynamic Framework provides a cohesive physical model spanning from nanometric optical slits to massive celestial orbits. By adopting a viscoelastic vacuum governed by an internal mesh tension (γ_0), relativistic limits and quantum probabilities are exposed as deterministic fluid dynamics. The open-source availability of the DWE and Dissipative Gravitation simulators allows for immediate peer reproduction of these emergent phenomena.

References

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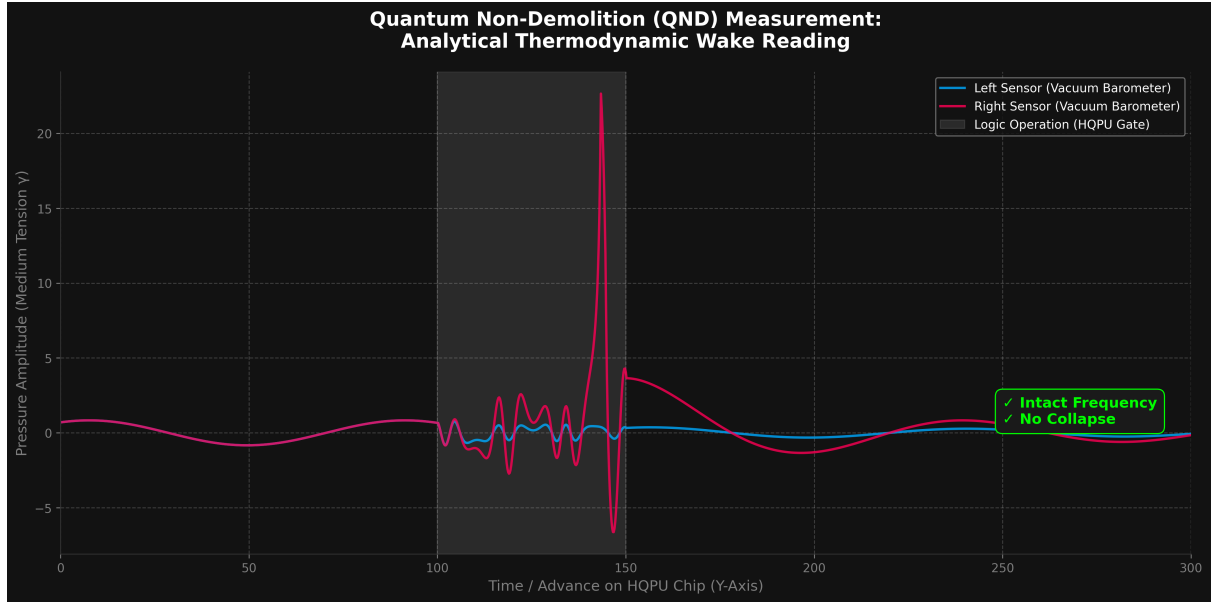


Figure 4: Quantum Non-Demolition (QND) measurement via thermodynamic wake barometry. The simulation demonstrates that information encoded in the vortex can be analytically read by lateral pressure sensors without damping the particle's internal helicity.

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